Day 1 | Control the AI



DECODING DATA SCIENCE

A prompt is a job description. Five parts turn a generic model into DecoBot, the DDS leave assistant.

THE FIVE-PART PROMPT RECIPE

01	02	03	04	05
Role	**Scope**	**Rules**	**Format**	**Facts**
Who is the AI?	What does it do, and refuse?	How does it act when unsure?	How does it answer?	What can it rely on?
DecoBot, the DDS leave assistant	Leave only. Salary and medical go to People & Culture	Never guess. Ask one clarifying question	Four short lines, direct answer first	A cheat sheet in the prompt. It stops scaling, so Day 2

EXPERIMENT

Same question, three prompts. Watch the behaviour change.

CLARITY LAB

Vague versus clear questions. Make the bot ask, not guess.

PROMPT CLINIC

Your turn: improve DecoBot until it passes five tests.

Day 2 | Teach DecoBot the Policy



DECODING DATA SCIENCE

Retrieval quality decides answer quality. We cut the policy three ways and measure which finds the right rule.

RAG IN FIVE STEPS

01
Split
Cut the PDF into chunks

02
Embed
Turn each chunk into numbers

03
Search
Find the chunks closest to the question

04
Answer
Gemini answers from those chunks only

05
Cite
Name the section, or say not found

THE CHUNKING LAB: THREE WAYS TO CUT THE SAME PDF

A | BIG CHUNKS
4 chunks
About 1000 tokens each. Rules mix with unrelated text.

B | SMALL CHUNKS
28 chunks
About 120 tokens. The 1-2 day notice line lands in another chunk.

C | BY SECTION
13 chunks
One chunk per numbered section. The notice table stays whole.

Day 3 | From Chatbot to Agent



DECODING DATA SCIENCE

Rules live in the policy, personal numbers live in the database. An agent picks the right tool, or both.

THREE READ-ONLY TOOLS

TOOL 1 | SQL

`get_leave_balance`

Annual and sick balance for one employee, read from the HR database.

TOOL 2 | SQL

`list_leave_requests`

Past, upcoming and pending requests for one employee.

TOOL 3 | RAG

`search_leave_policy`

The Day 2 policy search, wrapped as a tool. Answers rules, not balances.

ONE QUESTION, TWO TOOLS

JOHN DOE ASKS

I would like 10 working days off from 7 December. Any problem?

THE AGENT CALLS BOTH

get_leave_balance: 17 days left
search_leave_policy: 10 or more days needs 30 calendar days notice

THE COMBINED ANSWER

Balance and notice both pass: 17 days left, and 40 days ahead against the 30 required.